

Comparison of Irish Type and MVT Deposits: Genetic Constraints and Deposit Characteristics

Madeline Stapleton

GEOL 462

Outline

- Introduction
- Discuss and compare the general information about each deposit type
 - Similarities and differences – age, locations, ore type/minerals, commodities and host rocks
- Discuss and compare specific characteristics
 - Similarities and differences – mineralization, mechanisms of formation, tectonic settings, fluids and genetic constraints
- Summary and Conclusion
- References

MVT Overview

- MVT = Mississippi Valley Type Deposits
- Account for 24% of the global Pb and Zn resources
- Ore – Sphalerite, galena, iron sulphides
- Gangue – Carbonates
- Commodities
 - Primary – Pb and Zn
 - 10:1 mass ratio of Zn:Pb
 - Secondary – Ag, Cu, In
- Trace constituents – Ge, Ga, Cd
- Median deposit parameters
 - 7.0 Mt ore, 1.9% Pb, 6.0% Zn, 0.23% Cu, 32.5 g/t of silver
- Many different deposit subtypes
 - Ex. Irish Type deposits

Location

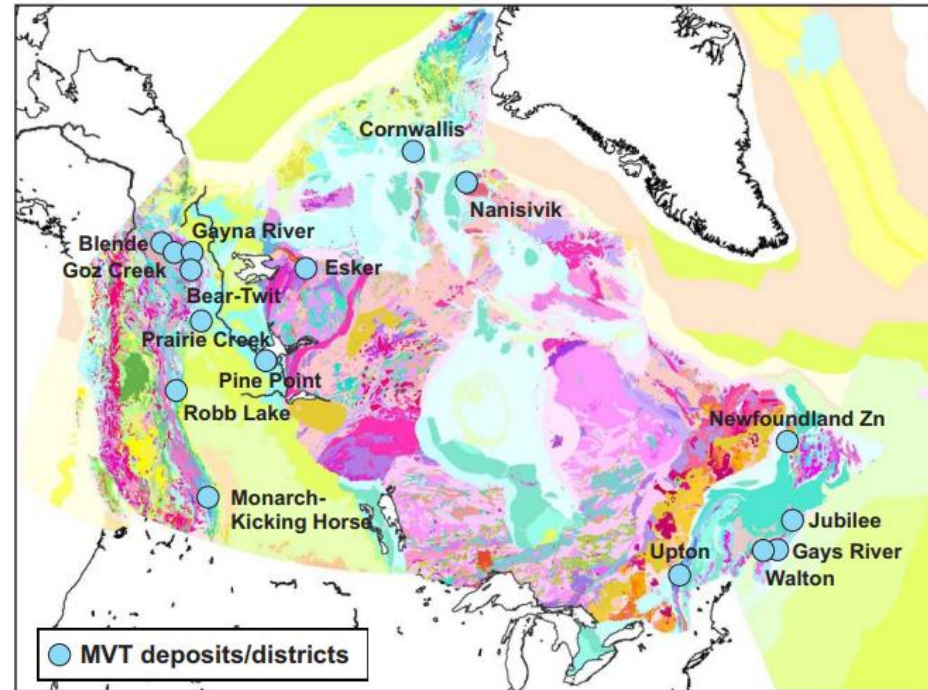
- Located all around the world
- Most intensely researched deposits are located in North America



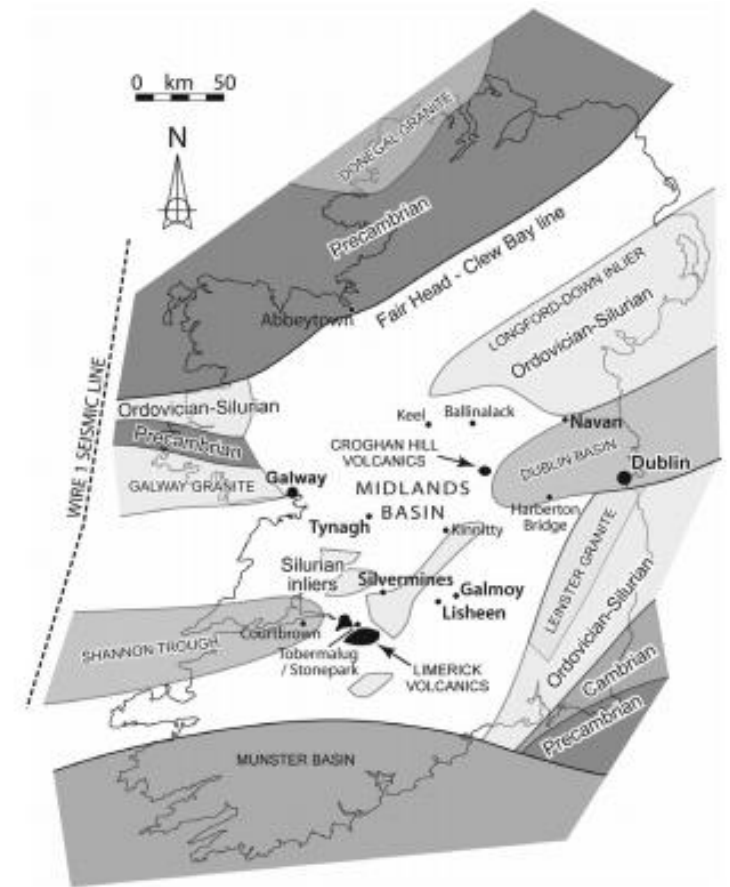
(Leach *et al.*, 2010)

Location

- Left image – distribution of the MVT deposits in Canada
- Right image – Locations of a variety of MVT deposits in the Irish Midlands



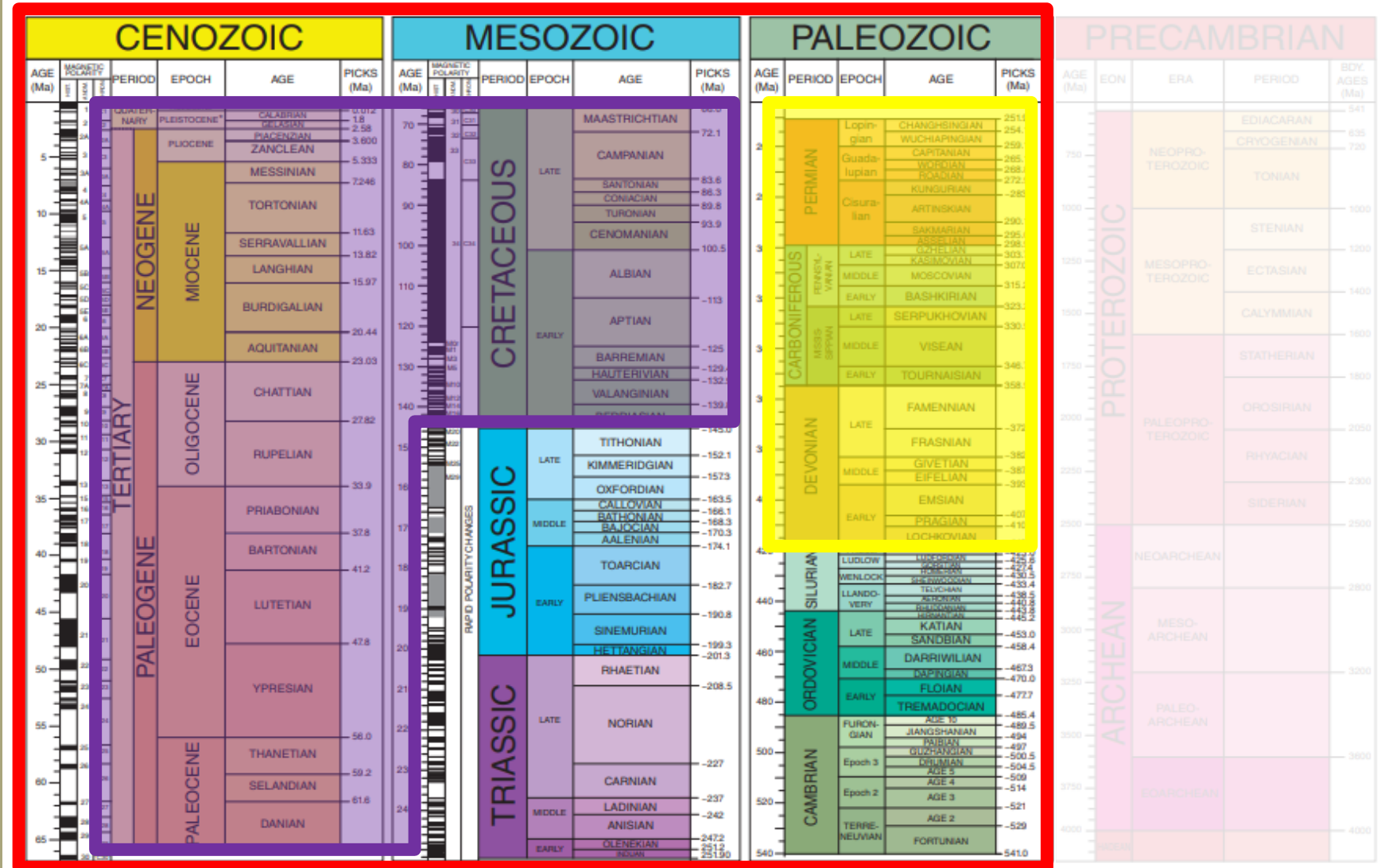
(Paradis *et al.*, 2007)



(Wilkinson and Hitzman, 2015)

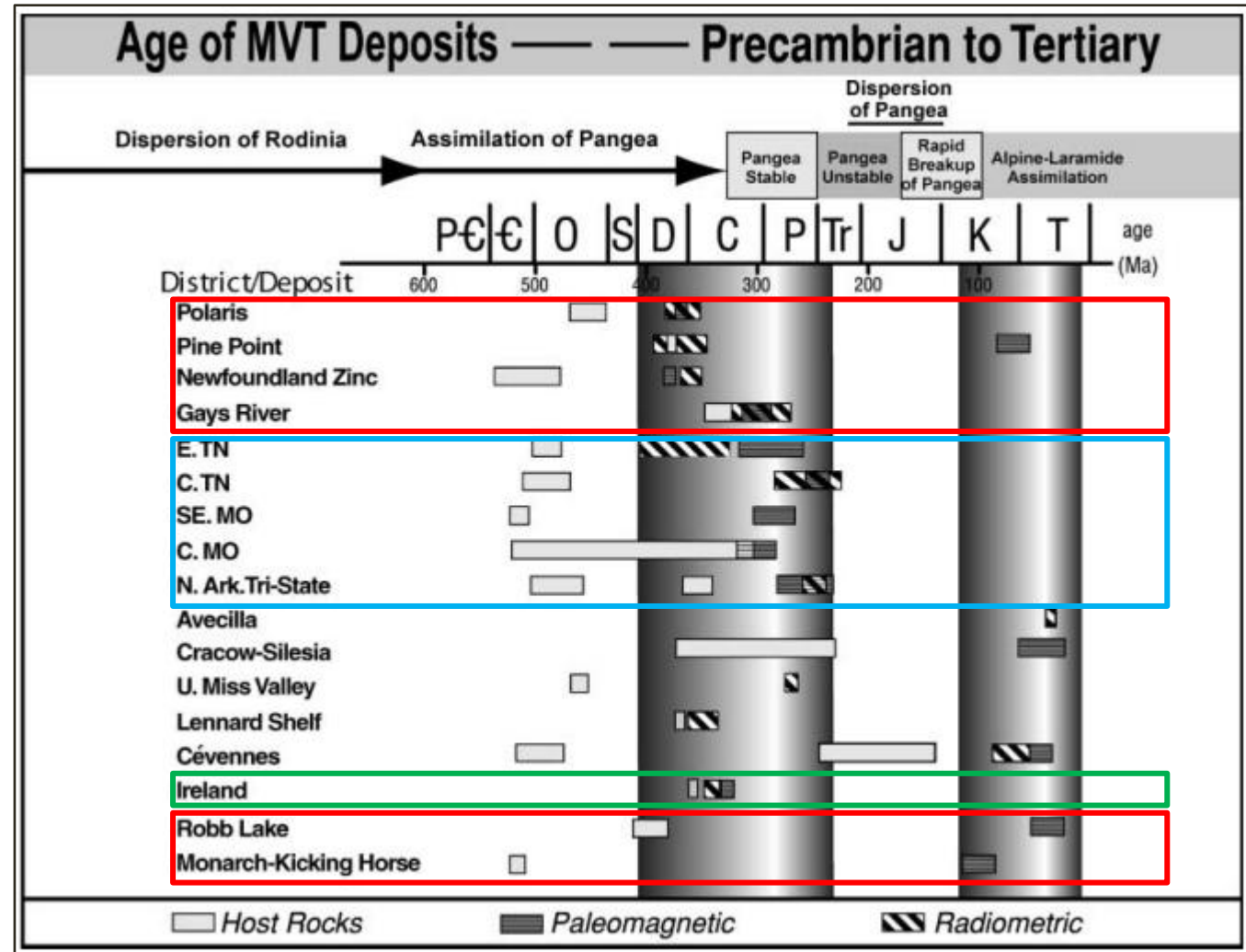
Overall Ages

- More than 80% of the deposits are in Phanerozoic rock
- Less than 20% are in Precambrian rocks
- Two main formation periods
 - Devonian to Permian
 - Cretaceous to Tertiary



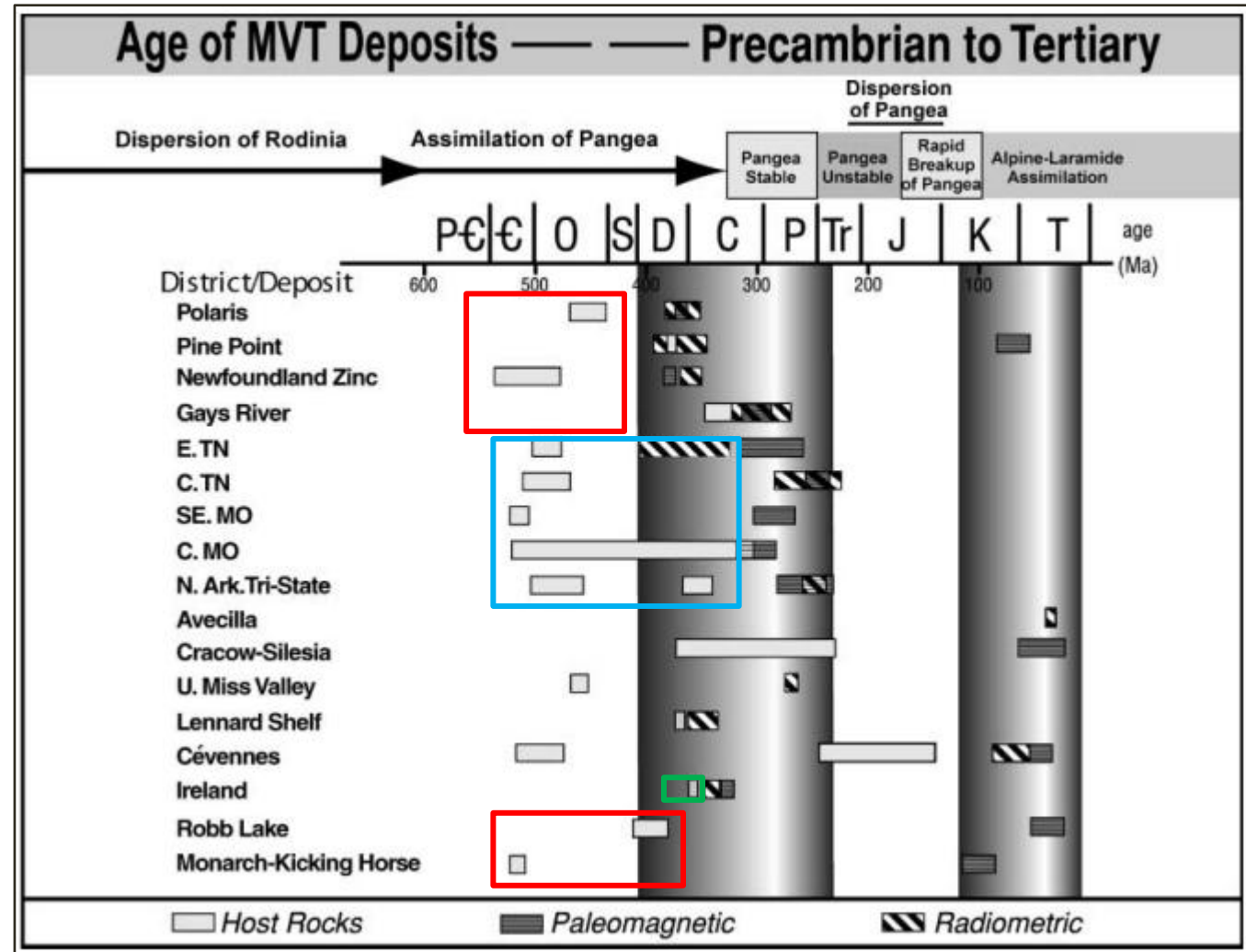
(Walker *et al.*, 2018)

Ages of the North American and Irish Type Deposits



(Leach *et al.*, 2001)

Host Rock Ages of the North American and Irish Type Deposits

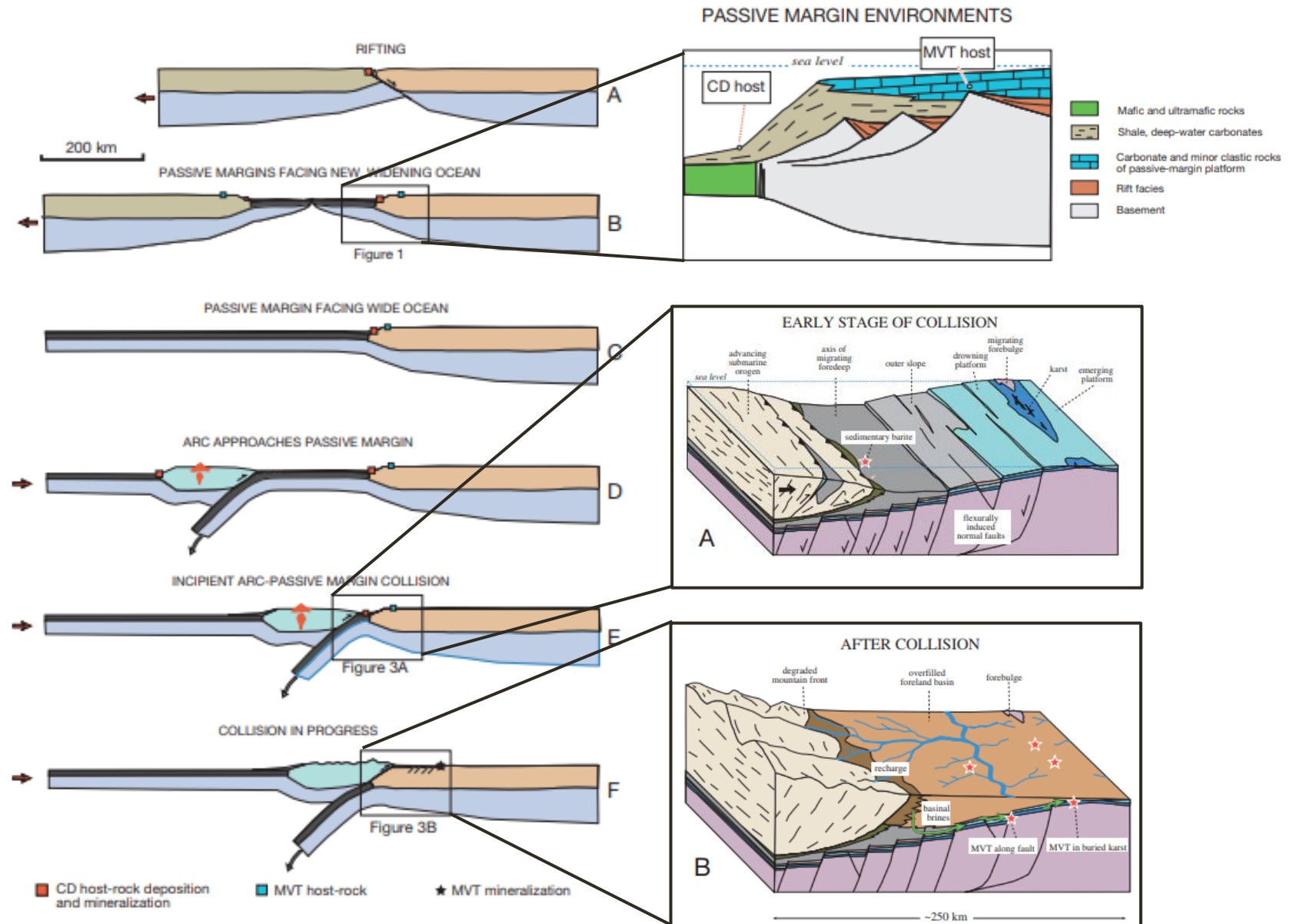


(Leach *et al.*, 2001)

Host Rock Composition

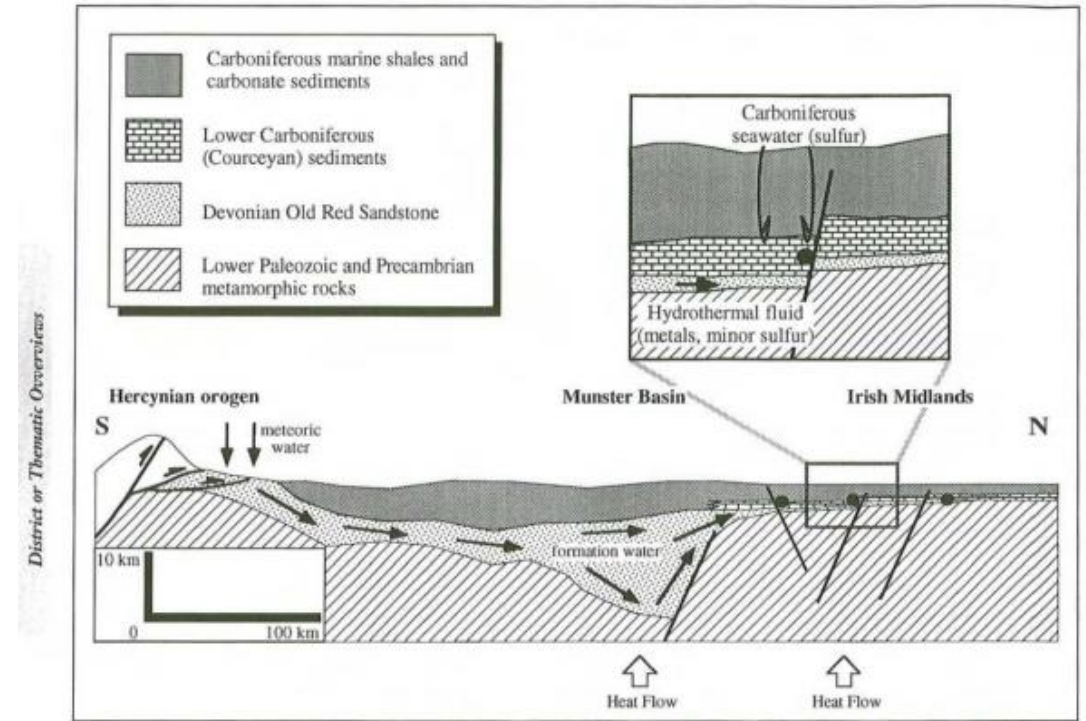
- Carbonates – medium to coarse grained
- White sparry dolomite will replace the original dolomite or limestone
- Mainly hosted in dolostone rather than limestone
 - May be due to the high transmissivity of fluid flow through dolostone compared to limestone
- The dolostone hosted deposits are generally larger with higher grades of Pb, Zn and Ag
 - One exception – the Navan deposit in North Central Ireland, it is a large high grade limestone hosted deposit
- Need a receptive host rock to form these deposits

Tectonic Setting and MVT Formation



Tectonic Setting and Irish Type Formation

1. Uplift in the highlands started fluid movement
2. Fluid movement through a confined aquifer
3. Change in tectonic regime causing the temperature of the fluids to rise
4. Fluids travelled to the carbonates and were focused in faults
5. Ore fluid mixed with seawater
6. Mineral Precipitation



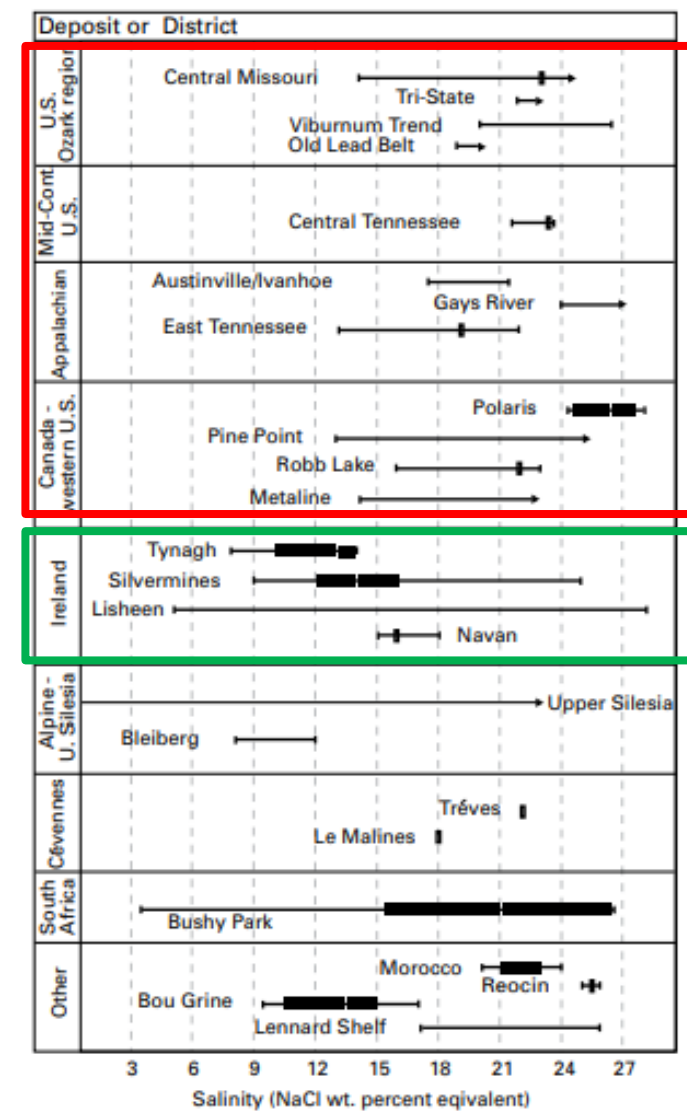
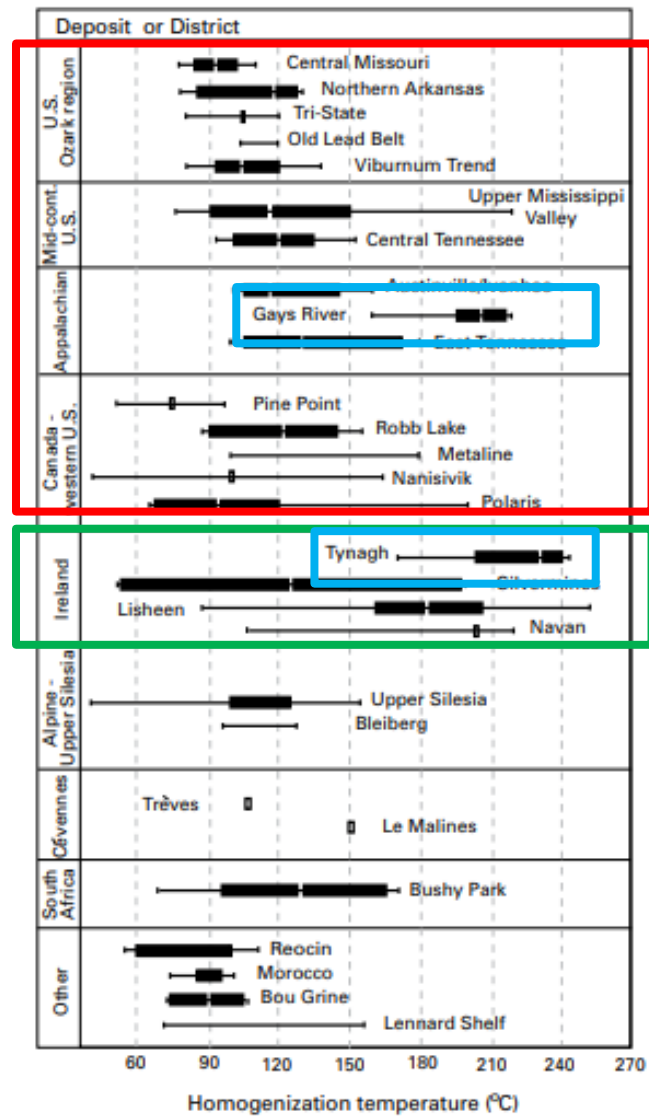
(Hitzman and Beaty, 1996)

Fluid Types

	MVT	Irish Type
Fluid composition	• Brine	• Moderate Salinity • Metal bearing
Metal Source	• Crustal rock interactions	• Wall rock interactions
Fluid Origin	• Saline basinal metalliferous fluids • Evaporated sea water	• Partially evaporated sea water
Fluid mixing	• Yes	• Yes
Metal transport Method	• Chlorine Complex	• Chlorine complex
Deposition/Precipitation	• Changes in T and pH	• Changes in T and pH

Fluids Properties

- Salinity
- Homogenization temperature
- pH



(Leach *et al.*, 2010)

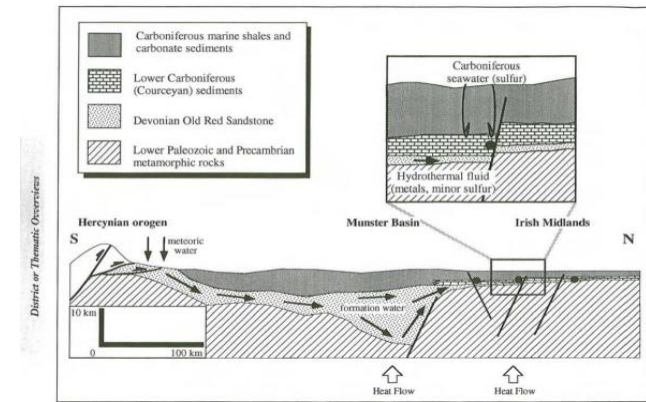
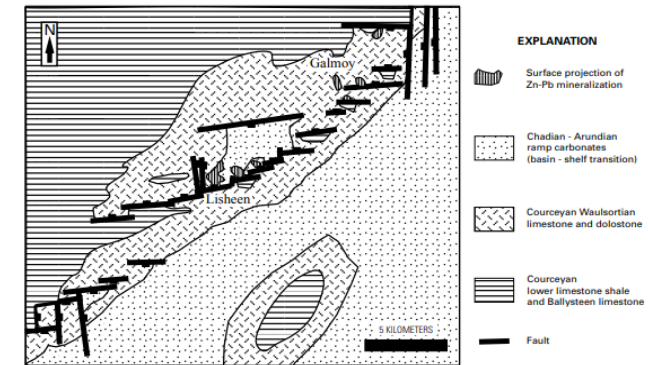
Controls on Fluid Flow and Ore Precipitation

- Carbonate dissolution collapse breccias
- Interconnected paleokarsts
- Carbonate and barrier Reefs
- Facies transitions
- Faults

DEPOSIT/ DISTRICT	SHALE EDGES	LIMESTONE/ DOLOSTONE	REEF AND BARRIER COMPLEX	SOLUTION COLLAPSE BRECCIAS	BASEMENT TOPOGRAPHY	FAULTS AND FRACTURES	PRE-ORE DISSOLUTION FEATURES
Viburnum Trend	●	●	●	●	●	●	
Old Lead Belt	●	●	●	●	●	●	
Tri-State	●	●	●	●	●	●	
North-Arkansas	●	●	●	●	●	●	
Cent. Tennessee		●		●		●	●
Mascot-Jefferson City (East Tenn.)		●		●		●	
Austinville	●			●		●	
Gays River			●		●	●	
Newfoundland Zn		●		●		●	●
Nanisivik				●		●	
Polaris	●	●		●		●	
Pine Point	●	●	●	●		●	●
Gayna				●		●	
Upper Mississippi Valley		●		●		●	
Alpine	●	●	●	●		●	
Robb Lake	●	●		●		●	
Monarch-Kicking Horse	●	●		●		●	
Upper Silesia	●	●		●	●	●	●
Lennard Shelf	●	●		●	●	●	●
Coxco				●		●	
Irish Midlands		●		●		●	●
Metaline	●	●		●		●	●
Cévennes		●		●	●	●	●
Bushy Park				●		●	●
Pering				●		●	●
Reocin				●		●	●

● Major ● Less important ● Minor

Ore controls in selected MVT districts

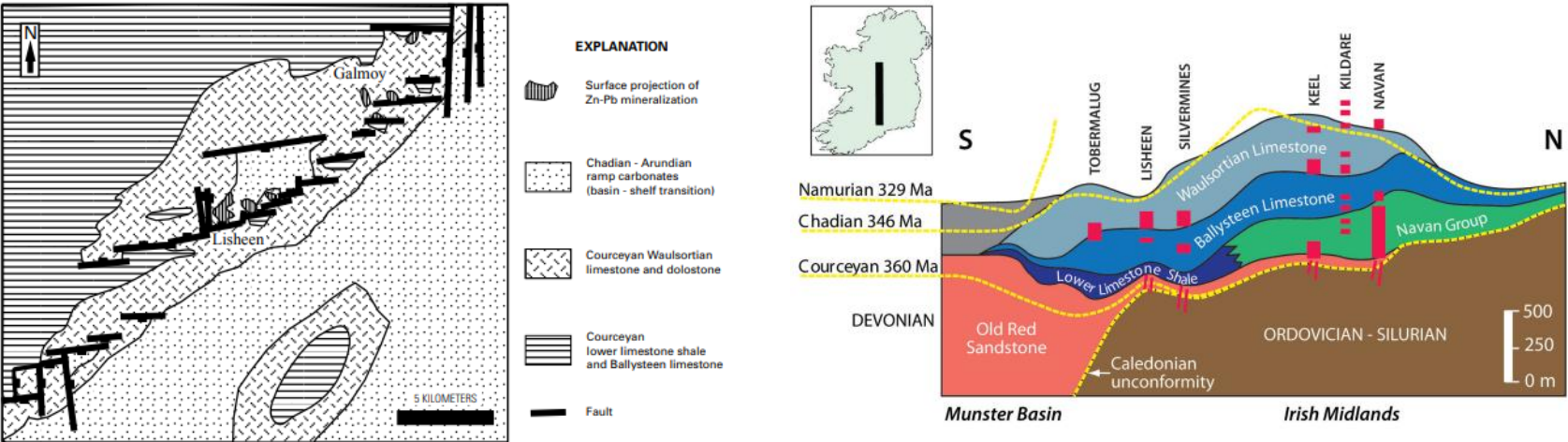


(Leach *et al.* 2010, Hitzman and Beaty, 1996 and Paradis *et al.*, 2007)

Mechanisms for Fluid Movement

- Topography
- Sediment Compaction
- Orogenic Compression
- Over pressured gas reserves
- Thermal and density reflux drives

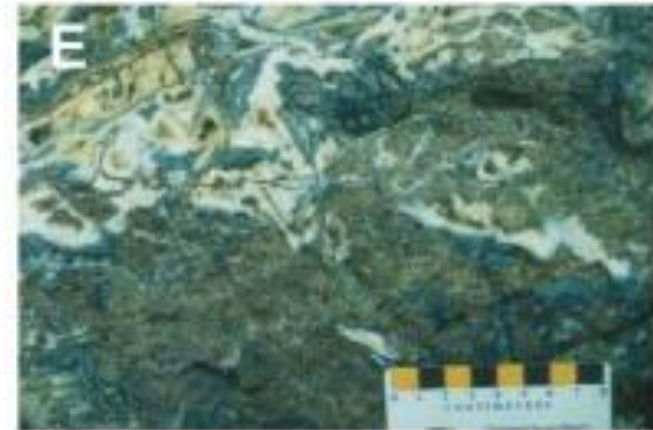
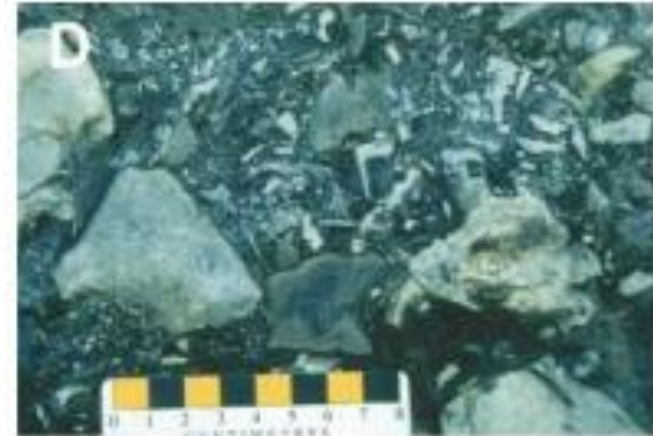
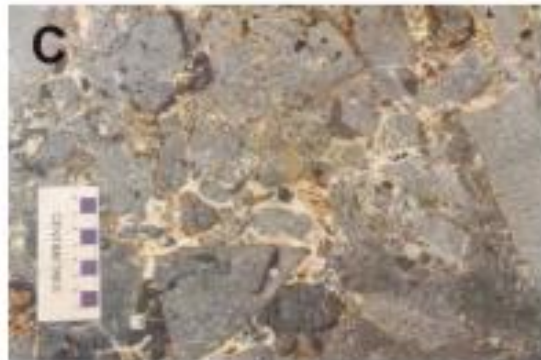
Mineralization



Deposit/District	Age of Host Rocks	Age of Mineralization					Orogeny
		Period	Epoch/Age	Date (Ma)	Method	Reference	
Polaris	Late - Middle Ordovician	Late Devonian	Famennian	367 $\pm$ 7	Paleomagnetism	Symons and Sangster (1992)	Ellesmerian
		Late Devonian	Famennian	366 $\pm$ 15	Sphalerite Rb-Sr	Christensen et al. (1995)	Ellesmerian
Nanisivik	Middle Proterozoic (Neohelikian)	Ordovician	Middle Ordovician	461	Ar-Ar on adularia	Sherlock et al. (2003)	?
		Proterozoic	Middle Proterozoic	1095 $\pm$ 10	Paleomagnetism	Symons et al. (2000)	Rifting
Gays River	Early Mississippian (Late Tournaisian - Early Viséan)	Pennsylvanian-Permian		297 $\pm$ 27	Ar-Ar on biotite	Kontak et al. (1994)	Alleghenian
		Pennsylvanian		303 $\pm$ 17	Paleomagnetism	Pan et al. (1993)	Alleghenian
Newfoundland Zn	Early Ordovician	Middle Devonian	Givetian	380 $\pm$ 7	Paleomagnetism	Pan and Symons (1993)	Acadian
		Late Devonian-Early Mississippian		360 $\pm$ 10	Ar-Ar on authigenic feldspar	Hall et al. (1989)	Acadian
Monarch-Kicking Horse	Middle Cambrian	Cretaceous	Late Cretaceous	100 $\pm$ 12	Paleomagnetism	Symons et al. (1998)	Laramide
Pine Point	Middle Devonian (Late Eifelian-Late Givetian)	Late Devonian-Early Carboniferous		361 $\pm$ 13	Sphalerite Rb-Sr	Nakai et al. (1993)	Antler
		Late Devonian-Early Carboniferous		374 $\pm$ 21	Rb/Sr isochron	Brannon et al. (1995)	Antler
		Late Devonian-Early Carboniferous		362 $\pm$ 9	Rb/Sr isochron on sphalerite + leachate	Nakai et al. (1993)	Antler
		Late Devonian-Early Carboniferous		71 $\pm$ 13	Paleomagnetism	Symons et al. (1993)	Laramide
Robb Lake	Late Silurian - Middle Devonian	Tertiary/Paleogene	Eocene / Lutetian	47 $\pm$ 10	Paleomagnetism	Smethurst et al. (1999)	Laramide
		Paleozoic		no good isochron; but similar to Pine Point at 360	Sphalerite Rb-Sr	Nelson et al. (2002)	Antler

(Leach *et al.* 2010, Wilkinson and Hitzman, 2015 and Paradis *et al.*, 2007)

Zoning and Alteration

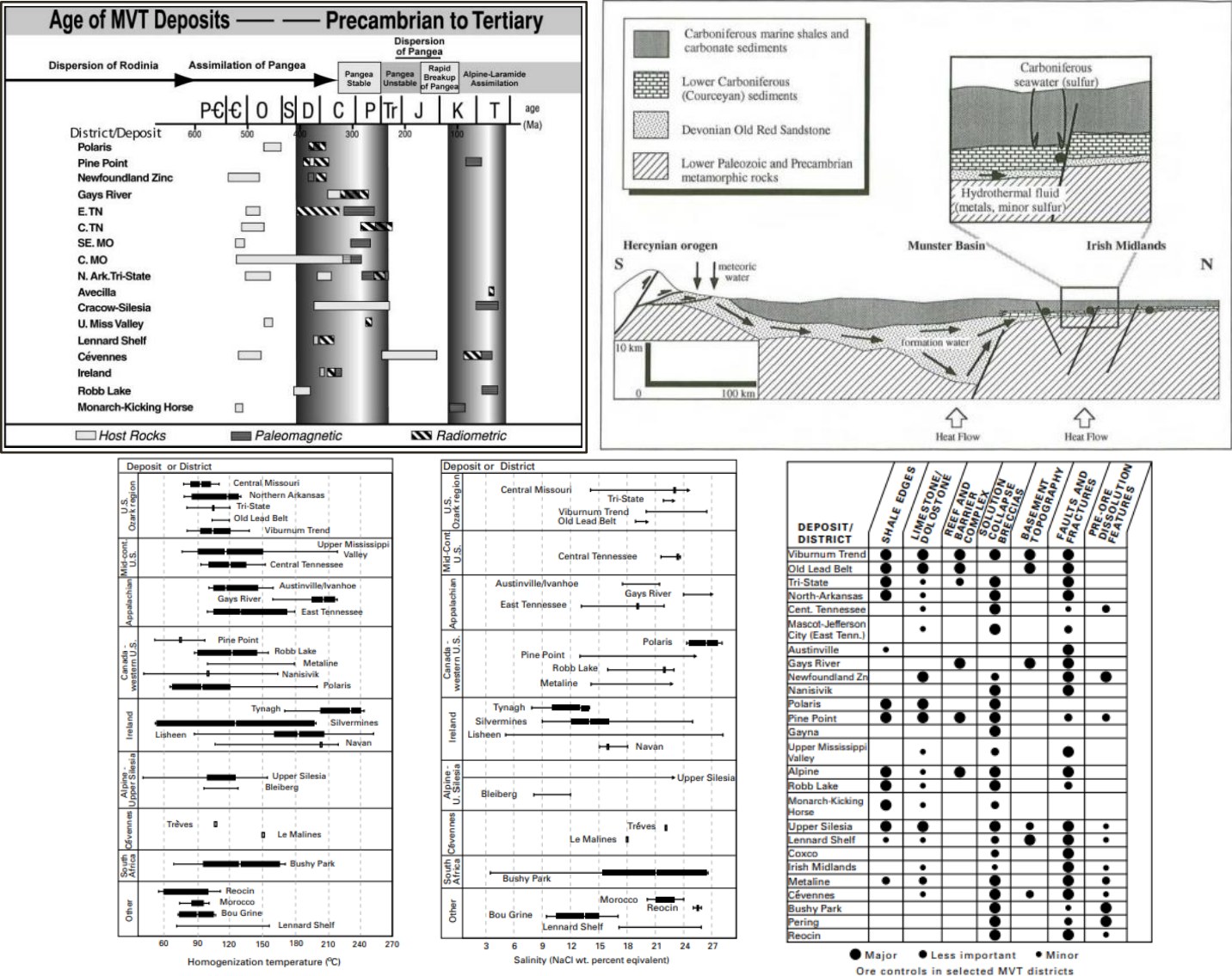


(Paradis *et al.*, 2007)

Genetic constraints

- Genetic constraints – features of the deposits which limit their formation, evolution and alteration
- Age ranges
- Fluid composition
- Host rocks
- Mechanisms for fluid transport

Genetic constraints



(Leach *et al.*, 2010)
(Hitzman and Beaty, 1996)

Summary and Conclusions

- Account for 24% of the global Pb and Zn resources
- Ore – Sphalerite, galena, iron sulphides
- Median deposit parameters – 7.0 Mt ore, 1.9% Pb, 6.0% Zn, 0.23% Cu, 32.5 g/t of silver
- Many different deposit subtypes – Irish Type deposits
- Carbonate hosted



(Leach *et al.*, 2010)

References

- Hitzman, M.W. and Beaty, D.W., (1996) The Irish Zn-Pb(-Ba) orefield, in Sangster, D.F., ed., Carbonate-hosted lead-zinc deposits: *Society of Economic Geologists, Special Publication Number 4*, p. 112-143.
- Kesler, S.E. and Carrigan, C.W., (2002) Discussion on “Mississippi Valley-type lead-zinc deposits through geologic time: implications from recent age-dating research” by D.L Leach, D. Bradley, M.T. Lewchuk, D.T.A Symons, G. de Marsily and J. Brannon (2001). *Mineralium Deposita*. 37. 800-802
- Leach, D.L., Taylor, R.D., Fey, D.L, Diehl, S.F., Saltus, R.W. (2010) A deposit model for Mississippi Valley Type lead-zinc ores, chap. A of Mineral deposit models for resource assessment. *U.S. Geological Survey Scientific Investigations Report 2010-5070-A*, 52 p.
- Leach, D.L., Bradley, D., Lewchuk, M.T., Symons, D.T., de Marsily, G., Brannon, J. (2001) Mississippi Valley-type lead-zinc deposits through geological time: implications from recent age-dating research. *Mineralium Deposita*. 36(8). 711-740
- Leach, D.L, Bradley, D.C., Huston, D., Pisarevsky, S.A., Taylor, R.D and Gardoll, S.J., (2010) Sediment-Hosted Lead Zinc Deposits in Earth History. *Society of Economic Geology*. 105.593 – 625
- Paradis, S., Hannigan, P. and Dewing, K., (2007) Mississippi Valley-type lead-zinc deposits (MVT). *Geological Association of Canada Mineral Deposit Division*. Special Publication n.5. 185-203
- Various Authors (1966). Genesis of Stratiform Lead-Zinc-Barite-Fluorite Deposits. New York, N.Y. The economic Geology Publishing Company
- Walker, J.D., Geissman, J.W., Bowring, S.A., and Babcock, L.E., compilers, 2018, Geologic Time Scale v. 5.0: Geological Society of America, <https://doi.org/10.1130/2018.CTS005R3C>. ©2018 The Geological Society of America
- Wilkinson, J.J, and Hitzman, M.W. (2015) The Irish Zn-Pb Orefield: The View from 2014. *Irish Association for Economic Geology*. p.59-72